

# Autonomous and Mobile Robotics

## Wheeled Mobile Robots: Mechanics, Constraints and Mobility Detailed lecture notes

Part I: slides of Prof. Giuseppe Oriolo; Part II: handwritten lecture notes  
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#### About these notes

These notes have two parts. **Part I** is based on the lecture slides (`WheeledMobileRobots1.Slides.pdf`, slide ranges in parentheses). **Part II** is based on the second half of the lecture, provided as handwritten notes (`AMR02.pdf`, page ranges in parentheses). No companion book chapter was provided for either part, so the formal statements (Definitions, Propositions, Lemmas, Theorems, proofs) formalise the content of the slides/notes with standard results. Whatever is *not* in the source is explicitly tagged (*beyond the slides*) or (*beyond the notes*). In Part II I corrected a typo in the notes ( $\partial h/\partial k$  read as  $\partial h/\partial q$ ) and added the solution of the example left open on page 6. Red dashed or framed boxes marked “Blank” are spots for photographs/videos on the slides that were deliberately not redrawn; all other figures are original TikZ drawings.

## Part I

# Mechanics of Mobile Robots

## 1 Outline of this lecture (*slides 1–2*)

The lecture opens the part of the course on *wheeled mobile robots* (WMRs). It covers four topics:

- **ground locomotion:** how a robot moves by contact with the ground;
- **balance:** what it takes for the robot not to fall;
- **wheels:** the three basic wheel types;
- **kinematic structures:** the classical ways of combining wheels on a chassis.

#### Intuition

Think of this lecture as the *hardware vocabulary* of the course. Later lectures write equations of motion (kinematics, control) for each structure; to do so one first needs to know which

wheel is where, which wheels are motorised, and which can steer. Everything here is about that bookkeeping.

## 2 Ground locomotion (*slide 3*)

Moving on the ground requires *contact* between the robot and the terrain. The slides distinguish two ways of establishing it.

**Definition 2.1** (Wheeled mobile robot). A *wheeled mobile robot* (WMR) typically consists of a single rigid body (the *base* or *chassis*) plus a set of wheels that connect it to the ground.

**Definition 2.2** (Legged robot). A *legged robot* typically consists of *several* rigid bodies articulated through joints; the contact with the ground is made through the feet.

Some robots achieve ground locomotion with neither wheels nor feet, for instance *snake robots*.

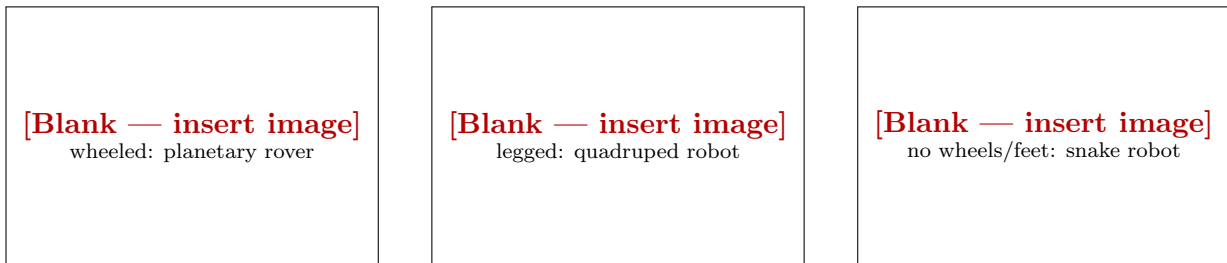


Figure 1: Photographs on slide 3 (not redrawn).

### Intuition

The difference is in the *mechanical complexity*: a WMR is one rigid body plus wheels, so its configuration is just the position and orientation of the chassis (plus steering angles); a legged robot is an articulated system, and every joint adds a degree of freedom to control. This is why the course starts with WMRs.

## 3 Balance (not falling) (*slides 4–5*)

### 3.1 Static balance and the support polygon

**Definition 3.1** (Support polygon). The *support polygon* is the convex hull of the points where the robot touches the ground: the wheel contact points for a WMR, the footprints for a legged robot.

**Definition 3.2** (Static balance). A robot is *statically balanced* when the vertical projection of its Center of Mass (CoM) on the ground falls inside the support polygon.

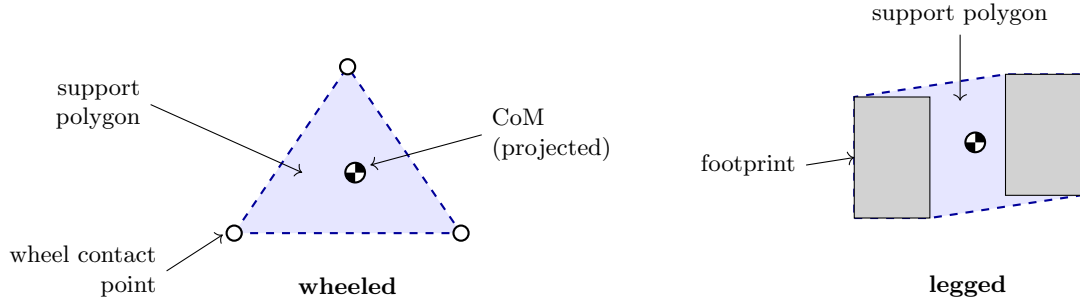


Figure 2: Support polygons (slide 4). Wheeled robot: triangle through three wheel contact points. Legged robot: convex hull of the two footprints. The projected CoM ( $\otimes$ -like symbol) must lie inside.

For WMRs the slides state a consequence: *one needs 3 wheels*. Here is the formal reason.

**Proposition 3.3** (Three wheels are necessary). *A robot whose support polygon has non-empty interior needs at least three ground contact points that are not collinear. Consequently, a WMR that must be statically balanced with a robust margin needs at least 3 wheels.*

*Proof.* The convex hull of one point is a point; of two points, a segment; of three or more collinear points, still a segment. All of these have empty interior in the plane, so the CoM projection can lie in them only by an unstable coincidence. Three non-collinear points span a triangle, which has non-empty interior.  $\square$

#### Intuition

A stool with one or two legs falls; a three-legged stool stands. With two wheels the robot can rotate about the line through the two contact points (like a seesaw), unless extra dynamic control is used. This is why the differential-drive robot of slide 9 adds a third, passive wheel (a caster) “for balance”.

The support polygon also tells us how the weight is shared among the wheels.

**Example 3.4.** A differential-drive robot has driving wheels at  $p_1 = (0, 0.3)$ ,  $p_2 = (0, -0.3)$  and a caster at  $p_3 = (0.4, 0)$  (metres), so  $A(p_1, p_2, p_3) = 0.12 \text{ m}^2$ . With  $c = (0.1, 0.05)$  the algorithm gives  $\lambda = (0.458, 0.292, 0.25)$ : all positive, so the robot is balanced and the caster carries 25% of the weight. After loading a payload that moves the CoM to  $c = (0.3, 0.1)$ , we get  $\lambda = (7/24, -1/24, 3/4)$ . Since  $\lambda_2 < 0$ , wheel 2 lifts and the robot tips over the edge  $p_1 p_3$ .

#### Intuition

The weight splits among the wheels like a seesaw: the closer the CoM is to a wheel, the more load that wheel carries. In the example the caster carries a quarter of the weight, so 75% of the weight presses on the driving wheels, which is what gives them traction. If the load moves far from the centre, one wheel is relieved of all weight and the robot tips. With four or more wheels the loads are no longer unique (like a four-legged table that wobbles on uneven ground), which is why real robots with more than three wheels need suspensions.

### 3.2 Dynamic balance and the Zero Moment Point (*slides 4–5*)

**Definition 3.5** (Dynamic balance). In *dynamic balance*, the CoM in the static criterion is replaced by the *Zero Moment Point* (ZMP), the point of the ground about which the horizontal components of the moment of gravity and inertial forces vanish. The robot is dynamically balanced when the ZMP lies inside the support polygon.

**Example 3.6.** With  $z_c = 0.8\text{ m}$ ,  $\ddot{z}_c = 0$  and  $\ddot{x}_c = 1\text{ m/s}^2$ :  $x_{\text{zmp}} = x_c - 0.8/9.81 \approx x_c - 8.2\text{ cm}$ . The ZMP lies 8.2 cm behind the CoM projection.

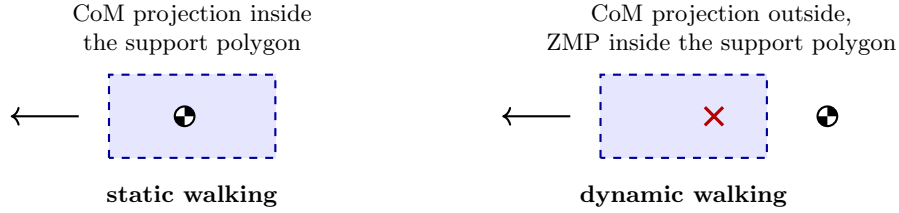


Figure 3: Static vs. dynamic walking (slide 5), shown for a single stance foot, top view. Arrows: walking direction. Red cross: ZMP.

#### Intuition

Carrying a tray while walking fast: you do not keep the tray's weight right above your feet, you lean and let motion help. Static walking is the cautious version, the CoM never leaves the foot area, so the robot could stop at any instant without falling; it is slow. In dynamic walking the CoM may be momentarily outside the foot, and the robot stays up because the accelerations move the ZMP back inside. The slide contrasts two humanoids walking one way or the other.

## 4 Wheels: 3 basic types (*slides 6–8*)

Wheels are classified by how their orientation relates to the chassis and by whether they are motorised (*active*) or free (*passive*).

### 4.1 Fixed wheel (*slide 6*)

**Definition 4.1** (Fixed wheel). A wheel with *fixed orientation* with respect to the chassis. It may be *active* (used for driving) or *passive* (used for balance).

### 4.2 Orientable (steerable) wheel (*slide 7*)

**Definition 4.2** (Orientable wheel). A wheel whose orientation with respect to the chassis is *variable*. It is typically *active* (used for steering).

### 4.3 Caster wheel (*slide 8*)

**Definition 4.3** (Caster wheel). A wheel with variable orientation with respect to the chassis, typically *passive* (used for balance) but possibly active, that *automatically aligns with the direction of motion*.

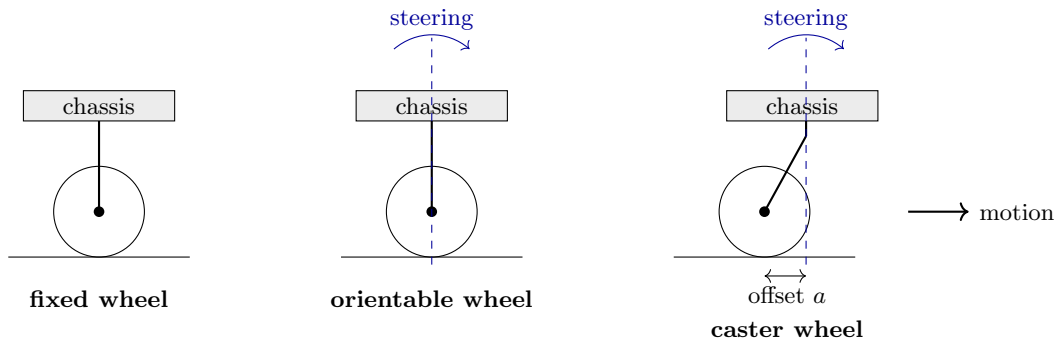


Figure 4: Side views of the three wheel types. The steering axis of the orientable wheel passes through the wheel centre; that of the caster is offset by a distance  $a$ , with the wheel trailing behind it.

| Wheel      | Orientation w.r.t. chassis | Typical role                          |
|------------|----------------------------|---------------------------------------|
| Fixed      | constant                   | active (driving) or passive (balance) |
| Orientable | variable                   | active (steering)                     |
| Caster     | variable, self-aligning    | passive (balance), possibly active    |

Table 1: The three basic wheel types (slides 6–8).

#### Intuition

It is the same reason a shopping-cart wheel swings round when you start pushing, or why a trailer follows the car: the wheel is *pulled* by a pivot ahead of it, and a pulled object trails behind. With  $v = 0.5 \text{ m/s}$  and  $a = 5 \text{ cm}$  the time constant is  $a/v = 0.1 \text{ s}$ : the wheel aligns in about a tenth of a second. The *offset* is essential: with  $a = 0$  the wheel would be an ordinary orientable wheel, which stays wherever it is steered. Fixed wheels give grip and balance, orientable wheels give steering, casters give free support.

## 5 WMR kinematic structures (*slides 9–15*)

The wheel types are combined on a chassis in a few classical ways. The slides present seven structures.

### 5.1 Differential-drive mobile robot (*slide 9*)

**Definition 5.1** (Differential-drive WMR). A WMR with **2 fixed wheels** on a common axle, **independently actuated**, plus **1 caster** added for balance.

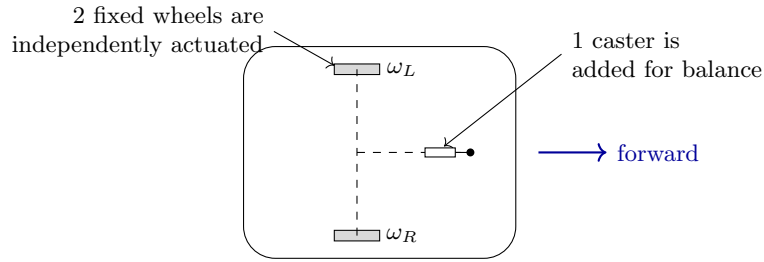


Figure 5: Differential-drive robot (slide 9), top view.

### Intuition

It is a tank (or a wheelchair): equal speeds drive straight, different speeds curve towards the slower wheel, opposite speeds spin on the spot. With  $r = 0.1$  m,  $d = 0.4$  m:  $\omega_R = \omega_L = 5$  rad/s gives  $v = 0.5$  m/s,  $\omega = 0$ ;  $\omega_R = -\omega_L = 5$  rad/s gives  $v = 0$ ,  $\omega = 2.5$  rad/s. The slides show an indoor mobile base and a robot vacuum cleaner as examples. The caster only keeps the robot from tipping, as the balance section requires.

## 5.2 Synchro-drive mobile robot (*slide 10*)

**Definition 5.2** (Synchro-drive WMR). A WMR with **3 orientable wheels** that are **simultaneously actuated**. The orientation of the chassis remains constant.

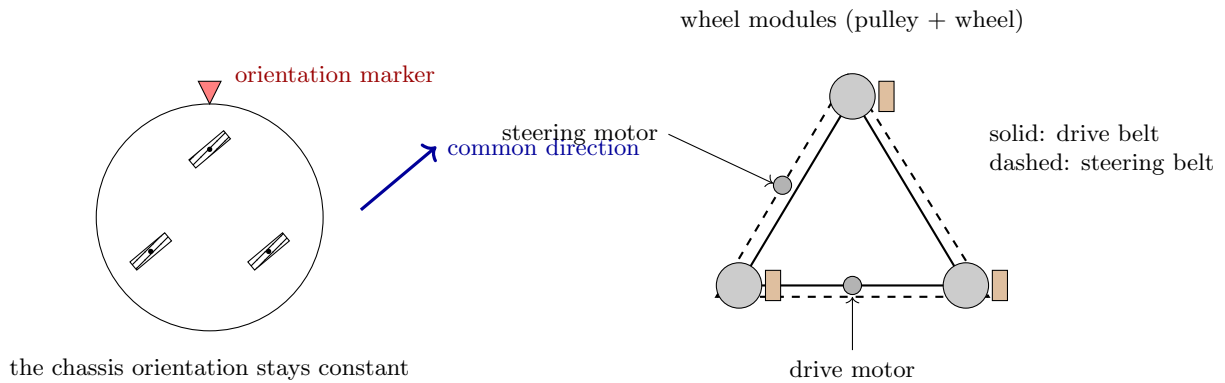


Figure 6: Synchro-drive (slide 10). Left: three orientable wheels, always parallel. Right: simplified drive train redrawn from the slide: one drive motor and one steering motor act on all the wheels through belts.

### Intuition

The three wheels are like three people pushing a table in exactly the same direction at the same speed: the table translates and never turns. The robot changes its direction of travel by turning all wheels together, but its body keeps pointing the same way; that is why the sensors or the “head” of a synchro-drive robot would need their own rotation mechanism (the slide shows a laptop riding on top). The price of this simplicity is a drive train with belts, but only two motors for three wheels.

### 5.3 Tricycle and car-like robots (*slide 11*)

**Definition 5.3** (Tricycle). **2 fixed wheels** on a common axle and **1 orientable wheel**.

**Definition 5.4** (Car-like). **2 fixed wheels** on a common axle and **2 orientable wheels** on a common axle.

Both may be front-wheel drive or rear-wheel drive.

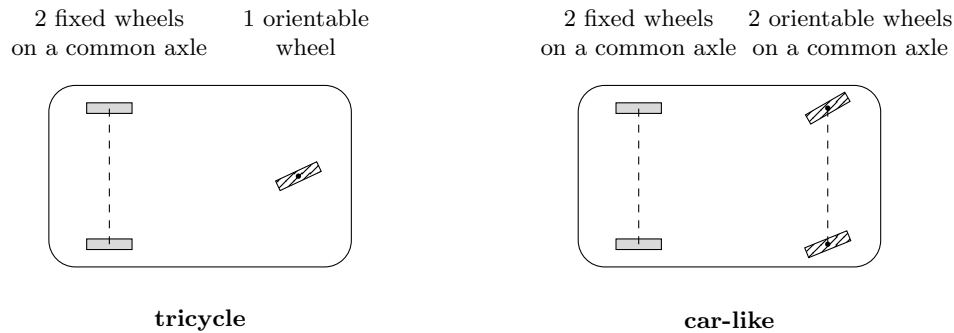


Figure 7: Tricycle (left) and car-like robot (right), top views (*slide 11*). The tilted wheels are steered.

#### Intuition

When a car turns, each wheel must roll along a circle with the same centre, and the inner circle is smaller; the inner wheel therefore has to turn more than the outer one. Numbers:  $\ell = 2.5\text{ m}$ ,  $w = 1.5\text{ m}$ ,  $\delta_i = 30^\circ$ :  $\cot \delta_o = 1.732 + 0.6 = 2.332$ , so  $\delta_o \approx 23.2^\circ$ . A tricycle, with only one steered wheel, does not have this issue. Front-wheel drive means the steered wheel(s) are also driven; rear-wheel drive means the fixed axle is driven, which requires the device of the next subsection when the two rear wheels share a motor.

### 5.4 The differential (*slide 12*)

**Definition 5.5** (Differential). A mechanical device needed whenever two *driving* wheels are mounted on a *common axle* (driven by the same motor), which allows the two wheels to move at *different speeds*.

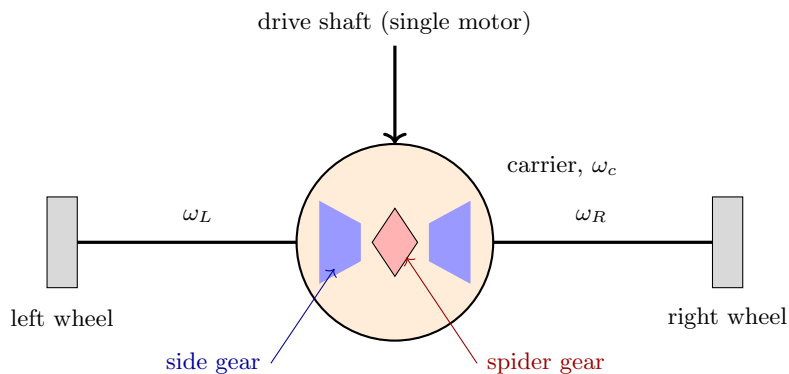


Figure 8: Schematic of a differential (idealised). The slide shows an animated 3D model.

### Intuition

Why is it needed? When a car turns, the outer wheel travels along a longer arc than the inner one, so the two must rotate at different speeds; if rigidly tied to one shaft they would drag and slip. The differential lets the wheels share the motor while spinning differently. Numbers: wheels at 4 and 6 rad/s  $\Rightarrow$  the carrier turns at 5 rad/s. A known side effect of equal torque split: if one wheel is on ice, it can only transmit a tiny torque, and so can the other.

## 5.5 Firetruck (*slide 13*)

**Definition 5.6** (Firetruck). A **tractor** (car-like) coupled to a **trailer** having **2 orientable wheels on a common axle**.

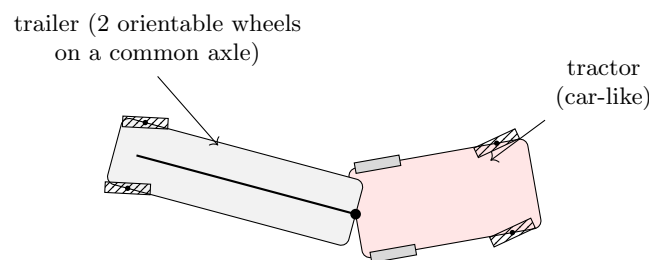


Figure 9: Firetruck (slide 13), top view; the dot is the hitch.

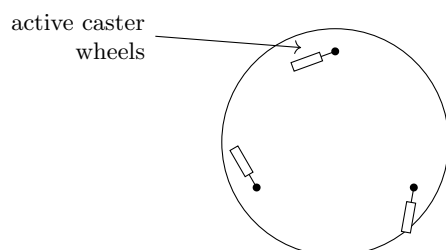
### Intuition

A ladder firetruck is very long. A second person (the “tiller operator”, as the slide’s overlay says) steers the *rear* axle so that the trailer does not cut the corner and can pass around cars and curbs. In kinematic terms, it is a car-like robot pulling a trailer with its own steering.

## 5.6 Omnidirectional robots with actuated caster wheels (*slide 14*)

**Definition 5.7** (Omnidirectional WMR). A WMR that can instantaneously move in any direction of the plane, *without first re-orienting the chassis*, and rotate independently of translation.)

The structure on the slide uses **active caster wheels**, i.e. caster wheels that are both steered and driven.



[Blank — insert image]  
actuated caster unit  
(steering motor + driving motor)  
and demo video on slide 14

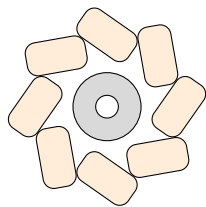
Figure 10: Omnidirectional robot with three active casters (slide 14, left drawing) and a spot for the slide’s photo/video.

### Intuition

Each wheel can point wherever it likes and be driven at the right speed, so it can follow the velocity that its own mount point must have. If all wheels point the same way, the robot translates in that direction; if their directions differ, the robot rotates. The effect is like a shopping cart that can instantly go sideways. So something that a car can't do, however the cost is mechanical: two motors per wheel. So more expensive, more complex and also more difficult to control than a differential-drive robot, but it can move in any direction without turning first.

## 5.7 Mecanum (Swedish) wheels (*slide 15*)

**Definition 5.8** (Mecanum wheel). A wheel with a ring of free-spinning rollers mounted around its rim, with the roller axes oblique to the wheel axle. Omnidirectional mobile robots can be built using Mecanum (Swedish) wheels.



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Mecanum robot (slide 15, top photo)

[Blank — insert image]

Mecanum wheelchair (slide 15, bottom photo)

Figure 11: A Mecanum wheel seen along its axle (slide 15, left drawing) and spots for the two photos.

### Intuition

The rollers turn the “wheel” into one that can roll forward *and* slide sideways freely: the ground force of each wheel acts along the axis of its roller instead of along the wheel plane. Choosing the speeds of several such wheels, one can obtain any combination of forward, sideways and rotational motion of the chassis, with no steering mechanism at all, a purely fixed-axle structure that is nonetheless omnidirectional. The problem is that the rollers have to be really close to the ground to transmit force, so the robot has to be very low to the ground. The slide shows a small robot and a wheelchair as examples.

## Part II

# Wheels: Constraints and Mobility

## 6 Wheels (constraints & mobility) (*page 1*)

Wheels do not merely support the robot: they *introduce constraints* on its motion. We can study them in four steps:

- **geometric constraints** (GC), which limit the configurations  $q$  the robot can take;
- **kinematic constraints** (KC), which limit the velocities  $\dot{q}$  it can have;
- the **relationships** between GC and KC;

- the resulting **mobility** of the robot,

and then apply everything to the example of **rolling wheels**.

### Intuition

Compare a bead on a wire with an ice skater. The bead can never leave the wire: a restriction on *where* it can be. The skater's blade cannot slide sideways at any instant: a restriction on *how* she can move, yet she can still reach every point of the rink. Both are “constraints” but of two different natures, and this part of the lecture is about telling them apart. Wheels turn out to be of the skater type.

## 7 Geometric constraints (GC) (pages 1–2)

### 7.1 Constraints on the configuration (page 1)

**Definition 7.1** (Configuration space). The configuration of the robot is a vector  $q \in \mathcal{C}$  of generalized coordinates, where the configuration space  $\mathcal{C}$  is (locally)  $\simeq \mathbb{R}^n$ . The number  $n$  is the number of coordinates.

Constraints are classified along two axes: *equality* (bilateral) vs. *inequality* (unilateral), and *time-variant* vs. *time-invariant*.

|                         | time-invariant | time-variant     |
|-------------------------|----------------|------------------|
| equality (bilateral)    | $h(q) = 0$     | $h(q, t) = 0$    |
| inequality (unilateral) | $h(q) \geq 0$  | $h(q, t) \geq 0$ |

Table 2: Classification of constraints on  $q$  (page 1). Only the first column of the first row is studied here.

**Definition 7.2** (Geometric constraint). Among the equality, time-invariant constraints, a *geometric constraint* (GC) has the form

$$h_i(q) = 0, \quad i = 1, \dots, k,$$

and *depends only on  $q$* .

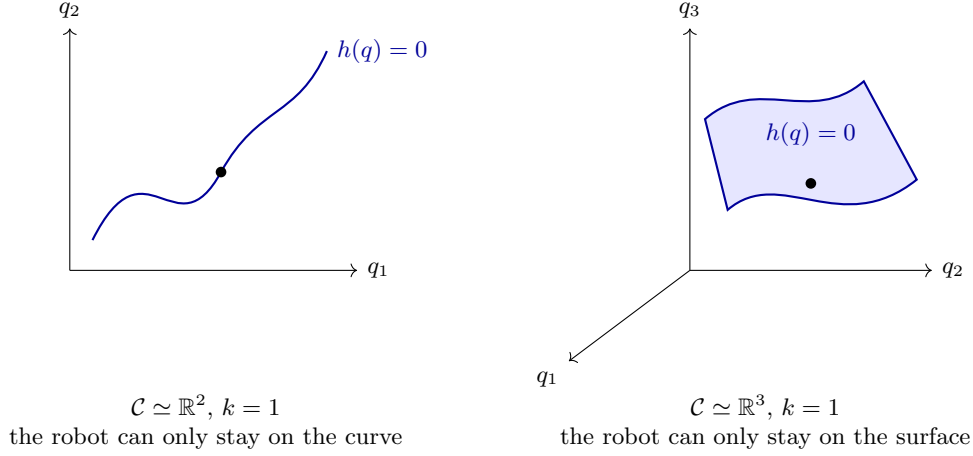


Figure 12: Geometric constraints (page 1): the dot is the robot, which must remain on the curve (left) or surface (right)  $h(q) = 0$ .

### Intuition

A bead threaded on a wire has  $n = 2$  coordinates in the plane but one constraint ( $k = 1$ ): it lives on a curve. A point constrained to a surface in space ( $n = 3$ ,  $k = 1$ ) lives on a surface. In general,  $k$  constraints leave an  $(n - k)$ -dimensional world for the robot to live in.

## 7.2 Mobility limitation and reduced coordinates (*page 2*)

**Proposition 7.3** (Global mobility limitation). *If the constraints  $h_1(q) = \dots = h_k(q) = 0$  are independent, i.e.  $\partial h / \partial q$  has rank  $k$ , the set  $\{q : h(q) = 0\}$  is (locally) an  $(n - k)$ -dimensional subset of  $\mathcal{C}$ , and the robot configuration must remain in it for all times.*

*Proof.* This is the implicit function theorem (regular value theorem): around any solution,  $k$  coordinates can be expressed as smooth functions of the remaining  $n - k$ .  $\square$

We call this a *global* mobility limitation: wherever the robot is, it is confined to a lower-dimensional subset.

### Intuition

Are we then “wasting” the  $k$  remaining coordinates? Yes: they are determined by the other  $n - k$ . The number  $n - k$  is the *number of degrees of freedom* (DOF), and one may try to describe the robot with a “reduced” configuration vector of  $n - k$  coordinates: solve the GC for  $k$  coordinates as functions of the independent ones and keep only the latter.

**Example 7.4** (The circle). Take  $q = (q_1, q_2) \in \mathbb{R}^2$  with  $q_1^2 + q_2^2 = 1$ . Solving for  $q_2$  gives  $q_2 = \pm\sqrt{1 - q_1^2}$ . This is only a *local* resolution: it needs a choice of branch, and at  $q_1 = \pm 1$  the derivative  $\partial h / \partial q_2 = 2q_2$  vanishes, so the square root breaks down (a *singularity*). So *we don't do that*. We use different coordinates instead, here the arc length  $s \in [0, 2\pi)$ , with  $q = (\cos s, \sin s)$ .

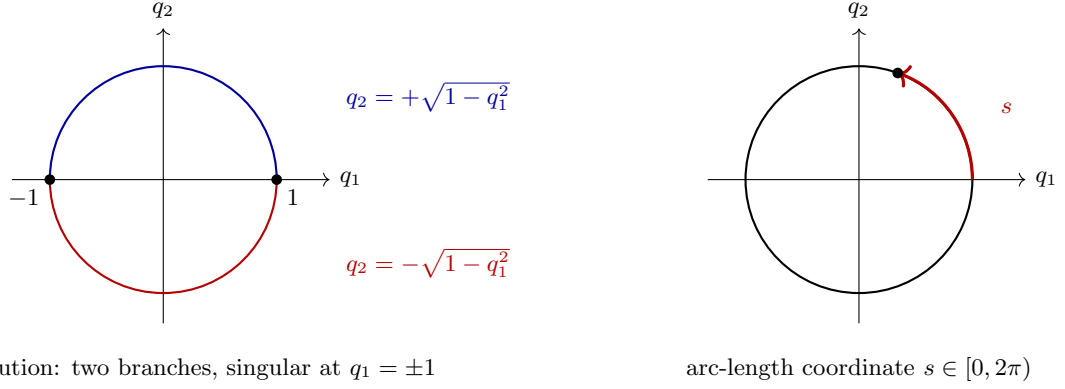


Figure 13: The circle  $q_1^2 + q_2^2 = 1$  (page 2): solving the GC for one coordinate versus using a single new coordinate  $s$ .

### Intuition

It is the difference between describing a point on the Earth's equator by “the latitude as a square root of something” and simply by its *longitude*: the longitude (our arc length  $s$ ) is one honest coordinate that wraps around, with no branches and no singular points. In general one picks new coordinates that automatically satisfy the GC, instead of solving the constraint equations.

## 8 Kinematic constraints (KC) (page 3)

**Definition 8.1** (Kinematic constraint). A *kinematic constraint* (KC) has the form

$$a_i(q, \dot{q}) = 0, \quad i = 1, \dots, k,$$

where  $\dot{q}$  are the *generalized velocities* (time derivatives). It depends on  $q$  and  $\dot{q}$ ; at any  $q$  it is a constraint on  $\dot{q}$ .

**Definition 8.2** (Pfaffian form). A KC is in *Pfaffian form* when  $\dot{q}$  appears *linearly*:

$$a_i^T(q) \dot{q} = 0, \quad i = 1, \dots, k \quad \Longleftrightarrow \quad A(q) \dot{q} = 0, \quad A(q) = \begin{bmatrix} a_1^T(q) \\ \vdots \\ a_k^T(q) \end{bmatrix}.$$

KC are typically found in this form.

**Proposition 8.3** (Admissible velocities). *At each configuration  $q$  the admissible generalized velocities are exactly those in the null space  $\mathcal{N}(A(q))$ . If  $A(q)$  has rank  $k$ , this is a linear subspace of dimension  $n - k$ .*

*Proof.*  $A(q)\dot{q} = 0$  is the definition of the null space, whose dimension is  $n - \text{rank } A(q) = n - k$  by rank-nullity.  $\square$

**Example 8.4.** For  $n = 3$  and the single constraint  $a^T = (2, -1, 0)$ , i.e.  $2\dot{q}_1 - \dot{q}_2 = 0$ , the admissible velocities are  $\dot{q} = \alpha(1, 2, 0)^T + \beta(0, 0, 1)^T$ : a plane ( $n - k = 2$  dimensions) in

velocity space.

### Intuition

The KC does not forbid any *place*; it forbids some *ways of moving* at each place: at every  $q$  the allowed velocities form a plane (or subspace), and the plane can change with  $q$ . This is a **local** mobility limitation. The number of instantaneous degrees of freedom is still  $n - k$ , but *we cannot “reduce” the configuration vector*: all  $n$  coordinates are still needed (whether this is really so is the subject of the next section).

## 9 Relationships between GC and KC (pages 4–6)

### 9.1 Any GC implies a KC (pages 4–5)

**Proposition 9.1** (GC  $\Rightarrow$  KC). *Every GC  $h(q) = 0$  implies the Pfaffian KC*

$$\dot{h}(q) = \frac{\partial h}{\partial q} \dot{q} = (\nabla_q h)^T \dot{q} = 0,$$

*i.e.  $a^T(q) = \partial h / \partial q$ . Hence admissible velocities belong to the tangent plane of the surface  $h(q) = 0$  at every  $q$ .*

*Proof.* For the constraint to hold as time goes by, it must hold continuously:  $h(q(t)) \equiv 0$ , so  $\dot{h} = 0$ . Since  $q$  depends on time, the chain rule gives  $\frac{\partial h}{\partial q} \dot{q} = 0$ , where  $(\frac{\partial h}{\partial q_1}, \dots, \frac{\partial h}{\partial q_n}) = (\nabla_q h)^T$  is the gradient. This is a KC in Pfaffian form.  $\square$

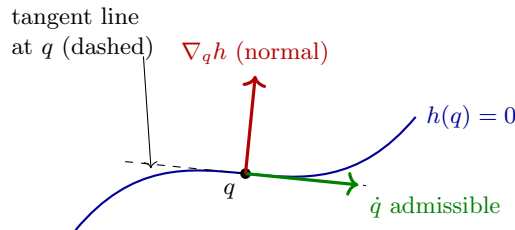


Figure 14: The gradient  $\nabla_q h$  is normal to the surface  $h(q) = 0$ ; admissible velocities are tangent to it, so  $(\nabla_q h)^T \dot{q} = 0$  (page 4).

### Intuition

Imagine hiking on a hillside: your velocity must be tangent to the slope, never pointing into the ground or up into the air. The gradient is the arrow sticking out of the surface (the normal), and “ $a^T \dot{q} = 0$ ” simply says that the velocity is *orthogonal to the normal*: a dot product between orthogonal vectors is zero. So every geometric constraint secretly hides a velocity constraint.

### 9.2 The converse: integrable vs. non-integrable (pages 5–6)

Can we say the contrary, i.e. that a KC  $a^T(q) \dot{q} = 0$  implies a GC  $h(q) = 0$ ? In general no, sometimes yes: it depends on whether the KC can be *integrated*.

**Definition 9.2** (Holonomic / nonholonomic). A Pfaffian KC  $a^T(q)\dot{q} = 0$  is *integrable* (holonomic) if there exist a function  $h(q)$  and a function  $\gamma(q) \neq 0$  such that

$$\frac{\partial h}{\partial q} = \gamma(q) a^T(q),$$

so that the KC is equivalent to  $\dot{h} = 0$ , i.e. to the GC  $h(q) = \text{const.}$  Otherwise it is *non-integrable* (nonholonomic), which is the general case.

**Example 9.3** (Integrable: circles). In  $\mathbb{R}^2$ ,  $q = (q_1, q_2)^T$ , take  $q_1\dot{q}_1 + q_2\dot{q}_2 = 0$ , i.e.  $a^T(q) = (q_1, q_2)$ . Consider  $q_1^2 + q_2^2 = C$  and differentiate:  $2q_1\dot{q}_1 + 2q_2\dot{q}_2 = 0$ ; dividing by 2 gives the KC. So it is *integrable*, with  $h = q_1^2 + q_2^2$  ( $\gamma = 2$ ) and  $C$  an integration constant.

**Proposition 9.4** (Local limitation  $\Rightarrow$  global limitation). *If a KC is integrable, every admissible trajectory stays on a level set of  $h$ :  $h(q(t)) = h(q(0))$  for all  $t$ .*

*Proof.*  $\frac{d}{dt}h(q(t)) = \frac{\partial h}{\partial q}\dot{q} = \gamma(q) a^T(q)\dot{q} = 0$ . □

**Example 9.5** (Left open by the professor:  $q_2\dot{q}_1 + q_1\dot{q}_2 = 0$  (solution by me, not in the notes)). Here  $a^T = (q_2, q_1)$  and  $h(q) = q_1q_2$  satisfies  $\partial h/\partial q = (q_2, q_1) = a^T$  (with  $\gamma = 1$ ): the KC is integrable and the robot is confined to a hyperbola  $q_1q_2 = c$ .

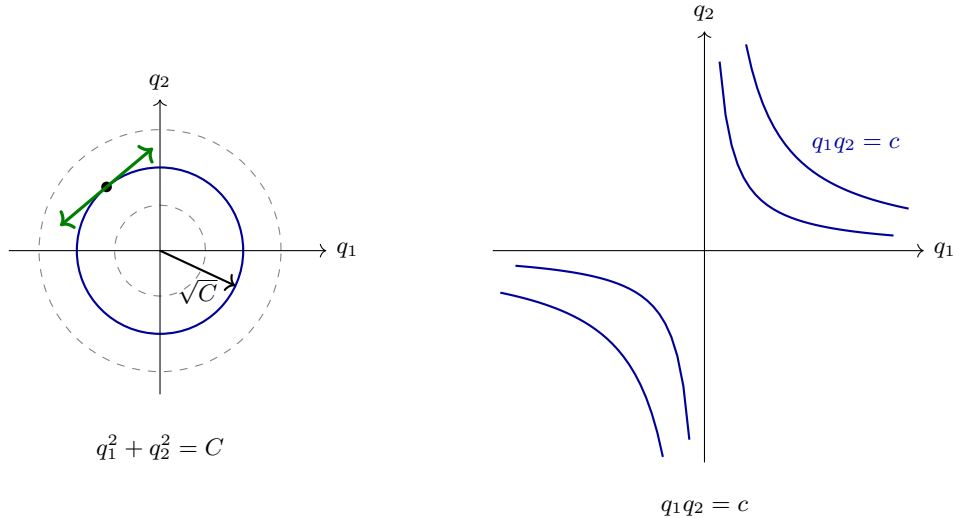


Figure 15: Left: the KC  $q_1\dot{q}_1 + q_2\dot{q}_2 = 0$  is tangent to circles: starting at a point we cannot leave the circle (page 6). Right: the KC  $q_2\dot{q}_1 + q_1\dot{q}_2 = 0$  integrates to hyperbolas.

### Intuition

At each point the admissible velocity is tangent to a curve; chaining these tiny tangent steps, you simply trace the curve out: you can never hop onto another circle. This is why we can say: *the local limitation (of the KC) implies a global limitation (of the GC)*. The function  $\gamma$  is just a rescaling: multiplying the constraint by a non-zero factor does not change which velocities are admissible. How does one know if a constraint is integrable without guessing  $h$ ? There are mathematical tools, such as **Frobenius' theorem**.

**Theorem 9.6** (Frobenius, single constraint in  $\mathbb{R}^3$  (stated without proof)). Let  $a(q) \neq 0$ ,  $q \in \mathbb{R}^3$ . The constraint  $a^T(q)\dot{q} = 0$  is integrable if and only if  $a \cdot (\nabla \times a) = 0$ .

## 10 Example: rolling wheels (pages 7–9)

### 10.1 The rolling disk and the pure rolling constraint (page 7)

If the KC is **not integrable** (nonholonomic), the mobility limitation is *purely local*. The key example is the rolling disk.

**Definition 10.1** (Rolling disk). A disk (wheel) that stays upright while rolling on the plane. Its configuration is  $q = (x, y, \theta)^T$ , where  $(x, y)$  is the contact point and  $\theta$  the orientation of the wheel plane (heading).

**Lemma 10.2** (Pure rolling constraint). Pure rolling without lateral slip means that the instantaneous velocity of the contact point has no component along the unit normal  $n = (\sin \theta, -\cos \theta)^T$  to the wheel plane:

$$\sin \theta \dot{x} - \cos \theta \dot{y} = 0, \quad \text{i.e.} \quad \underbrace{(\sin \theta, -\cos \theta)}_{n^T} \begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = 0.$$

In the full configuration space this is the Pfaffian KC

$$a^T(q)\dot{q} = (\sin \theta, -\cos \theta, 0) \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \end{pmatrix} = 0.$$

*Proof.* A velocity  $(\dot{x}, \dot{y})$  orthogonal to  $n$  is parallel to the heading  $(\cos \theta, \sin \theta)$ ; the formula is the orthogonality condition  $n^T(\dot{x}, \dot{y})^T = 0$ , with a zero entry for  $\dot{\theta}$  since the constraint does not involve it.  $\square$

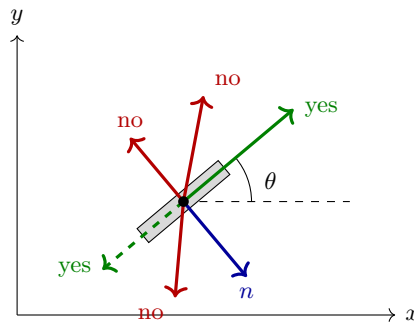


Figure 16: Top view of the rolling disk (page 7); the dot is the contact point  $(x, y)$ . Velocities of the contact point with a component along the normal  $n$  are forbidden (“no”); only motion along the wheel plane is allowed (“yes”).

#### Intuition

It is the ice-skater’s blade: the contact point can glide forwards or backwards along the blade, but can never slide sideways, so the velocity is always lined up with  $\theta$ . Since the condition

concerns *instantaneous* velocity, it is a *local* mobility constraint. In fact the null space of  $a^T(q)$  is spanned by  $g_1 = (\cos \theta, \sin \theta, 0)^T$  (roll) and  $g_2 = (0, 0, 1)^T$  (spin):  $\dot{q} = g_1 v + g_2 \omega$ , with  $n - k = 3 - 1 = 2$  admissible directions at each configuration.

## 10.2 Is pure rolling integrable? (page 8)

**Proposition 10.3** (Pure rolling is non-integrable). *The pure rolling constraint is non-integrable.*

We decide to avoid the use of Frobenius' theorem and prove it *constructively*, by proposing a maneuver that moves the disk from anywhere to everywhere: the *universal maneuver*.

**Algorithm 10.4** (Universal maneuver). Input: start  $q_s = (x_s, y_s, \theta_s)$  and goal  $q_g = (x_g, y_g, \theta_g)$ , with  $(x_g, y_g) \neq (x_s, y_s)$ . Let  $\varphi = \text{atan2}(y_g - y_s, x_g - x_s)$ .

1. **Pure rotation** about the vertical axis,  $\dot{x} = \dot{y} = 0$ , from  $\theta_s$  to  $\varphi$  (the KC is satisfied).
2. **Pure rolling**,  $\dot{q} = v(\cos \varphi, \sin \varphi, 0)^T$ , until  $(x_g, y_g)$  is reached (the KC is satisfied because the contact point velocity is always aligned with  $\theta = \varphi$ ).
3. **Pure rotation** from  $\varphi$  to  $\theta_g$  (the KC is satisfied).

(If  $(x_g, y_g) = (x_s, y_s)$ , only a rotation is needed.)

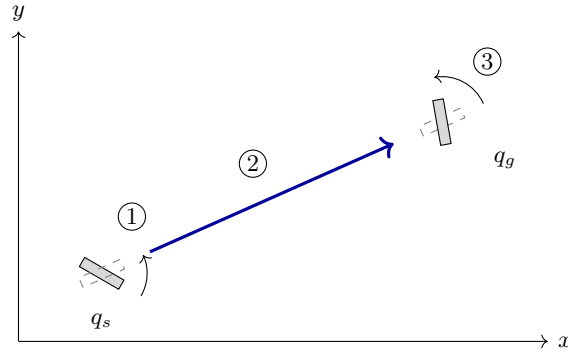


Figure 17: The universal maneuver (page 8): ① rotate in place towards the goal, ② roll straight to  $(x_g, y_g)$ , ③ rotate to the final orientation. Dashed: intermediate configurations.

**Proposition 10.5** (The maneuver is universal). *For any  $q_s, q_g \in \mathbb{R}^2 \times S^1$ , Algorithm 10.4 produces a path that satisfies the pure rolling constraint at every instant and connects  $q_s$  to  $q_g$ .*

*Proof.* During phases 1 and 3,  $\dot{x} = \dot{y} = 0$  so  $a^T \dot{q} = 0 \cdot \sin \theta - 0 \cdot \cos \theta + 0 \cdot \dot{\theta} = 0$ . During phase 2,  $\theta = \varphi$  is constant and  $\dot{q} = v(\cos \varphi, \sin \varphi, 0)$ , so  $a^T \dot{q} = v(\sin \varphi \cos \varphi - \cos \varphi \sin \varphi) = 0$ . The three phases end respectively at heading  $\varphi$ , at position  $(x_g, y_g)$  and at heading  $\theta_g$ .  $\square$

**Corollary 10.6** (Non-integrability). *The pure rolling constraint is non-holonomic.*

*Proof.* If it were integrable, by Proposition 9.4 every admissible trajectory would stay on a level set of a function  $h$  with  $\partial h / \partial q = \gamma a^T \neq 0$ , so  $h$  would not be constant; but by the universal maneuver 10.4 admissible paths connect any two configurations, so  $h$  would be constant on  $\mathbb{R}^2 \times S^1$ . Contradiction.  $\square$

**Example 10.7.** Let  $q_s = (0, 0, 0)$  and  $q_g = (1, 1, \pi/2)$  (metres, radians). Then  $\varphi = \pi/4$ : rotate by  $\pi/4$  in place, roll  $\sqrt{2} \approx 1.41$  m along the diagonal, then rotate by another  $\pi/4$  to reach  $\theta_g = \pi/2$ .

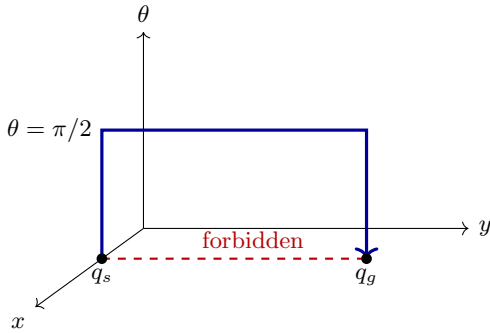
### Intuition

It is the recipe of “turn towards the goal, walk straight, turn to the final direction”. Each of the three motions is allowed by the KC, and together they reach *any* configuration, even though at every instant only 2 of the 3 directions of  $q$ -space are admissible. A constraint that could be integrated would trap the robot on a curve or surface, whereas here nothing traps it.

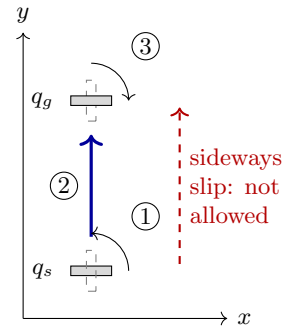
## 10.3 No global mobility limitation (page 9)

Hence there is **no global mobility limitation**: the KC *must be non-holonomic*. Here we illustrate this with a “naive representation” of  $\mathcal{C}$  and with parallel parking.

**Example 10.8** (Parallel parking of a rolling disk). Let  $q_s = (0, 0, 0)$  and  $q_g = (0, 2, 0)$ : the goal is 2 m to the side, with the same orientation. The direct motion  $\dot{y} \neq 0$  with  $\theta = 0$  is not admissible, since  $a^T \dot{q} = -\dot{y} \neq 0$ : parallel parking *cannot be done instantaneously*. It can be done with three admissible moves: (1) rotate to  $\theta = \pi/2$ ; (2) roll 2 m along  $y$  (at  $\theta = \pi/2$  the constraint reads  $\dot{x} = 0$ , so  $y$  is free); (3) rotate back to  $\theta = 0$ .



(a) path in  $\mathcal{C}$



(b) top view

Figure 18: Parallel parking (page 9). (a) In  $\mathcal{C} = \mathbb{R}^2 \times S^1$  the direct path at  $\theta = 0$  is forbidden; the admissible path climbs along the  $\theta$  axis, moves along  $y$ , and comes back down. (b) The same maneuver seen from above.

### Intuition

Think of the three coordinates  $(x, y, \theta)$  as a 3D world in which the disk may move only along two of the three directions at each point; yet by combining the motions, like a sailor tacking against the wind, every point becomes reachable. So the kinematic constraint *reduces how you can move* (local), but *not where you can get to* (global). Compare the two kinds of constraint: an integrable KC is a GC in disguise (global limitation, the dimension of the reachable set drops to  $n - k$ ); a non-integrable KC (rolling) keeps all  $n$  coordinates alive and limits only the instantaneous velocities.

# 11 Summary

## 11.1 Part I: kinematic structures of WMRs

| Structure            | Wheels                                                               | Key feature                                                       | Slide examples                    |
|----------------------|----------------------------------------------------------------------|-------------------------------------------------------------------|-----------------------------------|
| Differential-drive   | 2 fixed, independently actuated, + 1 caster (balance)                | Steering through the speed difference of the two wheels           | indoor base, robot vacuum cleaner |
| Synchro-drive        | 3 orientable, simultaneously actuated                                | Wheels always parallel; chassis orientation constant              | cylindrical robot with laptop     |
| Tricycle             | 2 fixed on a common axle + 1 orientable                              | Front- or rear-wheel drive                                        | —                                 |
| Car-like             | 2 fixed + 2 orientable, each pair on a common axle                   | Front- or rear-wheel drive; differential for a driven common axle | —                                 |
| Firetruck            | car-like tractor + trailer with 2 orientable wheels on a common axle | Trailer steered separately by the tiller operator                 | ladder truck                      |
| Omni, active casters | active caster wheels (steered and driven)                            | Any direction without re-orienting; 2 motors per wheel            | actuated caster unit              |
| Omni, Mecanum        | wheels with oblique free rollers                                     | Omnidirectional with fixed wheel axes                             | small robot, wheelchair           |

Table 3: Part I: comparison of the WMR kinematic structures (slides 9–15).

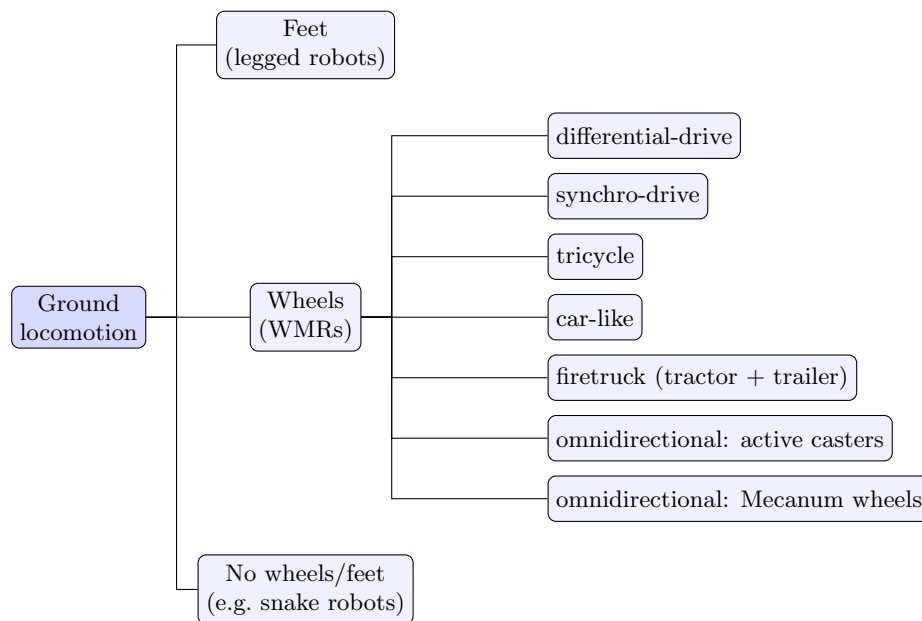


Figure 19: Part I map: from ground locomotion to the WMR kinematic structures.

## 11.2 Part II: constraints and mobility

| Constraint                               | Form                                                            | Mobility limitation                                                                                             | Example                                                      |
|------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| Geometric (GC)                           | $h(q) = 0$ , depends only on $q$                                | <i>Global</i> : $q$ confined to an $(n - k)$ -dimensional set; reduced coordinates possible                     | circle $q_1^2 + q_2^2 = 1$ , arc length $s$                  |
| Kinematic, integrable (holonomic)        | $a^T(q)\dot{q} = 0$ with $\partial h / \partial q = \gamma a^T$ | Equivalent to a GC: local limitation implies a global one                                                       | $q_1\dot{q}_1 + q_2\dot{q}_2 = 0$ : circles                  |
| Kinematic, non-integrable (nonholonomic) | $a^T(q)\dot{q} = 0$ , no such $h$                               | <i>Purely local</i> : $\dot{q} \in \mathcal{N}(A(q))$ ; every configuration is reachable; $q$ cannot be reduced | rolling disk $\sin \theta \dot{x} - \cos \theta \dot{y} = 0$ |

Table 4: Part II: types of constraints and their effect on mobility (pages 1–9).

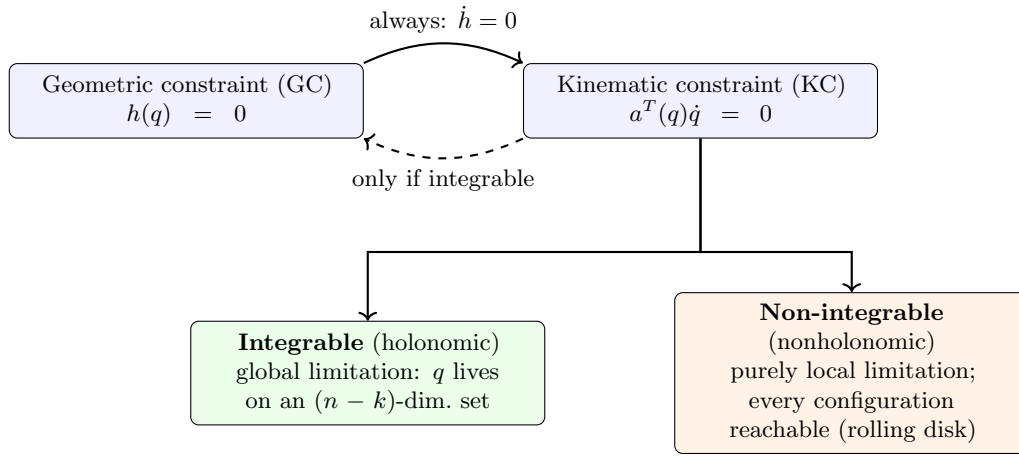


Figure 20: Part II map: relationships between geometric and kinematic constraints.

## 11.3 Key take-aways

- Ground locomotion needs contact (wheels or feet). Static balance requires the CoM projection inside the support polygon, hence at least 3 wheels; dynamic balance replaces the CoM with the ZMP.
- Wheels come in three types (fixed, orientable, caster) which combine into the classical structures: differential-drive, synchro-drive, tricycle, car-like, tractor-trailer, and omnidirectional robots (active casters or Mecanum wheels); a differential is needed when two driven wheels share an axle.
- Wheels introduce constraints on  $q$ . A geometric constraint  $h(q) = 0$  confines the robot globally to an  $(n - k)$ -dimensional set; a kinematic constraint  $a^T(q)\dot{q} = 0$  restricts the velocities locally to the null space of  $A(q)$ .
- Every GC implies a KC (the velocity must be tangent to the surface,  $\dot{h} = 0$ ); the converse holds only for integrable (holonomic) KC, for which the local limitation implies a global one.
- Pure rolling,  $\sin \theta \dot{x} - \cos \theta \dot{y} = 0$ , is non-integrable (Frobenius, or the universal “turn, roll, turn” maneuver): there is no global mobility limitation, all of  $(x, y, \theta)$  is reachable, but not instantaneously, as in parallel parking.